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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 2

Game Based Learning

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA04

SLOT :

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Duration: 1 Hour

Assessment Tool Weightage: 35%

Marks: 50

Code of Conduct:

I U . S o m e s h K u m a r (1 9 2 4 2 5 0 1 9) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

U. Somesh Kumar

Signature of the student:

U. Somesh Kumar

Name of the student:

Criteria	Understanding of Concept (2.5)	Conceptual Clarity (2.5)	Problem-Solving Ability (2)	Solution Design & Application (2)	Teamwork & Game Participation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Round 1						
Round 2						
Total						

QUESTIONS:01**Total Marks: 50****Question:**

Design and participate in a two-round game-based learning activity to explore the fundamental concepts of Convolutional Neural Networks (CNNs), including architecture, layers, filters, parameter sharing, regularization, AlexNet, ResNet, and real-world applications. Progress through five levels in each round, applying your knowledge to solve challenges involving CNN design, feature extraction, architecture comparison, and practical applications.

The Game Progression**Round 1:**

Basic CNN Concepts → Filters → Parameter Sharing → Architecture → AlexNet vs ResNet

Round 2:

Convolution → Feature Maps → Regularization → Architecture Comparison → Real-World Application

ROUND 1: CNN Architecture Builder (Learn by constructing and understanding CNN architectures layer by layer)

Level	Challenge	Description
Level 1	CNN Layer Matcher	Match CNN layers such as Convolution, Pooling, Flatten, and Fully Connected layers with their respective functions and roles in the network.
Level 2	Filter Finder	Identify suitable filters and kernels for feature extraction and understand how filter size, stride, and padding affect the output feature map.
Level 3	Parameter Sharing Puzzle	Determine how the same filter weights are reused across different image locations and calculate the number of parameters in a convolutional layer.
Level 4	CNN Architecture Debugger	Identify and correct problems in a CNN architecture, such as incorrect layer order, incompatible dimensions, and inappropriate layer selection.
Level 5	AlexNet vs ResNet Challenge	Compare AlexNet and ResNet architectures and identify how their architectural approaches differ, particularly the use of residual connections in ResNet.

ROUND 2: CNN Feature Detective (Learn by exploring how CNNs extract features, avoid overfitting, and solve real-world problems)

Level	Challenge	Description
Level 1	Edge Detection Explorer	Apply simple (3\times3) filters to an image and determine how convolution operations can detect edges and other basic features.

Level	Challenge	Description
Level 2	Feature Map Detective	Analyze feature maps produced by different CNN layers and identify the progression from simple features such as edges to complex patterns and objects.
Level 3	Regularization Rescuer	Identify overfitting in a CNN model and select suitable regularization techniques to improve generalization and model performance.
Level 4	CNN Architecture Race	Compare ResNet and AlexNet for different image-classification applications and determine which architecture is more suitable for a given problem.
Level 5	CNN Application Master	Design a CNN-based solution for a real-world application such as medical image analysis, face recognition, object detection, or autonomous driving, and justify the selected architecture and layers.

Game-Based Learning Activity on Convolutional Neural Networks (CNNs)

Introduction

Convolutional Neural Networks (CNNs) are one of the most important deep learning techniques used for processing and understanding visual data. CNNs are particularly effective for image classification, object detection, face recognition, medical image analysis, autonomous vehicles, security systems, and many other real-world applications. Unlike traditional image-processing methods, CNNs can automatically learn useful features from images, starting from simple patterns such as edges and gradually progressing toward complex shapes and objects.

To understand the fundamental concepts of CNNs effectively, a two-round game-based learning activity can be conducted. The activity combines theoretical knowledge with problem-solving challenges. Each round consists of five progressive levels. The first round focuses mainly on constructing and understanding CNN architectures, while the second round focuses on feature extraction, regularization, architecture comparison, and practical applications.

The game encourages students to learn concepts such as convolutional layers, pooling layers, filters, feature maps, parameter sharing, fully connected layers, regularization, AlexNet, and ResNet through interactive challenges. Each level increases in difficulty, allowing participants to apply the concepts learned in previous levels.

The two-round game is titled:

Round 1 – CNN Architecture Builder

Round 2 – CNN Feature Detective

The overall objective is to help learners understand not only what CNN components are, but also why they are used and how they work together to solve image-processing problems.

ROUND 1: CNN Architecture Builder

Objective of Round 1

The first round introduces the basic structure of CNNs. Participants act as CNN architects and progressively construct a neural network by solving five levels. Each level develops a specific concept and prepares the learner for the next challenge.

Level 1 – CNN Layer Matcher

Challenge

The first challenge requires participants to match different CNN layers with their corresponding functions.

The major layers considered are:

- Convolutional layer
- Pooling layer
- Flatten layer
- Fully connected layer

Learning Activity

Participants are given cards containing layer names and separate cards containing their functions. They must correctly match each layer with its role.

The convolutional layer applies learnable filters to the input image. It extracts important visual features such as edges, corners, textures, and shapes.

The pooling layer reduces the spatial dimensions of feature maps. Max pooling, for example, selects the maximum value from a small region. This reduces computation and makes the network less sensitive to small changes in image position.

The flatten layer converts a multidimensional feature map into a one-dimensional vector. This allows the extracted features to be passed into fully connected layers.

The fully connected layer uses the extracted features to perform classification or prediction. Each neuron is connected to neurons in the previous layer.

Expected Outcome

By completing this level, participants understand the purpose and sequence of the major CNN layers. They learn that a typical CNN follows a pattern such as:

Input Image → Convolution → Activation → Pooling → Convolution → Pooling → Flatten → Fully Connected → Output

A correct match earns points and allows the participant to proceed to Level 2.

Level 2 – Filter Finder

Challenge

In this level, participants learn how filters or kernels are used to extract features from an image.

A filter is a small matrix of numerical values that slides over an image and performs convolution. Common filter sizes include 3×3 and 5×5 .

Learning Activity

Participants are shown different filters and asked to identify what type of feature each filter can detect.

For example, a simple vertical edge filter can be represented as:

-1 0 1

-1 0 1

-1 0 1

A horizontal edge filter can be represented as:

-1 -1 -1

0 0 0

1 1 1

Participants apply these filters to small image matrices and observe the resulting feature maps.

The challenge also introduces three important convolution parameters:

Filter Size

The filter size determines the region of the image examined at one time. A 3×3 filter examines a small 3×3 region and is commonly used because it can capture local features efficiently.

Stride

Stride determines how many pixels the filter moves after each operation. A stride of 1 moves the filter one pixel at a time, while a stride of 2 moves it two pixels at a time.

Padding

Padding adds extra pixels around the boundary of an image. It helps preserve spatial dimensions and allows edge information to be processed.

The output size of a convolution operation can be calculated using:

$$\text{Output Size} = ((N - F + 2P) / S) + 1$$

where:

- N = input size
- F = filter size
- P = padding
- S = stride

Expected Outcome

Participants understand how filters extract features and how filter size, stride, and padding affect the output feature map.

Level 3 – Parameter Sharing Puzzle

Challenge

The third level introduces one of the most important concepts of CNNs: parameter sharing. Instead of using different weights for every location in an image, CNNs use the same filter weights repeatedly as the filter moves across the image.

Learning Activity

Participants are given an image and a 3×3 filter. They must identify the locations where the same filter is applied.

For example, if a convolutional layer receives an RGB image with three input channels and uses 10 filters of size 3×3, the number of parameters is:

$$3 \times 3 \times 3 \times 10 + 10 = 280$$

The additional 10 parameters represent the bias associated with each filter.

The general formula is:

Number of Parameters = Filter Height × Filter Width × Input Channels × Number of Filters + Number of Filters

Participants solve several parameter-counting problems and identify how parameter sharing reduces the number of learnable parameters compared with a fully connected network.

Importance of Parameter Sharing

Parameter sharing provides several advantages:

- Reduces the number of parameters.
- Reduces computational requirements.
- Helps the network detect the same feature at different locations.
- Improves learning efficiency.
- Provides translation-related robustness.

Expected Outcome

Participants understand why CNNs are computationally efficient and how the same filter can recognize a feature regardless of where it appears in an image.

Level 4 – CNN Architecture Debugger

Challenge

In this level, participants become CNN architecture "debuggers." They are provided with incorrect CNN architectures and must identify and correct the mistakes.

Learning Activity

Example problems may include:

Incorrect Architecture:

Input → Fully Connected → Convolution → Pooling → Output

The participant must identify that convolutional processing should normally occur before flattening and fully connected classification.

Another problem may involve incompatible dimensions. For example, a feature map may be incorrectly connected to a fully connected layer without applying a flattening operation.

Participants must determine:

- Whether the layer order is correct.
- Whether dimensions are compatible.
- Whether pooling is appropriate.
- Whether the network contains unnecessary layers.
- Whether the output layer matches the classification problem.

Expected Outcome

The activity helps participants understand that designing a CNN is not simply about adding more layers. Each layer must have a specific purpose, and its input and output dimensions must be compatible with neighboring layers.

Participants receive points for every correctly identified and fixed architectural problem.

Level 5 – AlexNet vs ResNet Challenge

Challenge

The final level of Round 1 compares two important CNN architectures: AlexNet and ResNet.

AlexNet

AlexNet was a landmark deep CNN architecture that demonstrated the effectiveness of deep learning for large-scale image classification. It contains convolutional and pooling layers followed by fully connected layers.

Important characteristics include:

- Deep convolutional architecture compared with earlier systems.
- Use of ReLU activation.
- Use of pooling layers.
- Use of dropout for reducing overfitting.
- Large-scale image classification capability.

ResNet

ResNet, or Residual Network, introduced residual connections or skip connections. These connections allow information to bypass one or more layers and directly reach deeper layers.

A simplified representation is:

Input → Convolution → Convolution → Addition with Input → Output

The residual connection helps address the degradation problem that can occur when neural networks become very deep.

Game Challenge

Participants are given statements describing AlexNet and ResNet. They must identify which architecture each statement belongs to.

For example:

- Uses residual/skip connections → ResNet
- Uses ReLU extensively → AlexNet
- Designed to support very deep networks → ResNet
- Landmark architecture in early large-scale deep learning → AlexNet

Round 1 Outcome

After completing all five levels, participants should understand CNN architecture, filters, parameter sharing, layer ordering, and major CNN architectures.

ROUND 2: CNN Feature Detective

Objective of Round 2

Round 2 moves from architecture construction to practical CNN reasoning. Participants become "feature detectives" who investigate how CNNs extract information from images, prevent overfitting, select architectures, and solve real-world problems.

Level 1 – Edge Detection Explorer

Challenge

Participants perform convolution manually using simple 3×3 filters.

Learning Activity

A small grayscale image represented as a matrix is provided. Participants apply an edge-detection kernel to different regions of the image.

For example:

-1 0 1

-1 0 1

-1 0 1

The filter responds strongly when there is a change in intensity from one side of the region to the other. Therefore, it can help identify vertical edges.

Participants calculate the convolution values and construct the resulting feature map.

Expected Outcome

The learner understands that convolution transforms raw image pixels into useful feature representations. Instead of directly understanding an entire image, the CNN initially detects basic visual patterns such as:

- Edges

- Lines
- Corners
- Simple textures

Level 2 – Feature Map Detective

Challenge

Participants analyze feature maps generated by different CNN layers and determine what type of information each layer has learned.

Learning Activity

The participant is shown feature maps from different depths of a CNN.

Early layers generally detect simple features such as:

- Edges
- Lines
- Corners
- Basic textures

Middle layers combine these features into:

- Shapes
- Patterns
- Object parts

Deeper layers identify more complex concepts such as:

- Faces
- Animals
- Vehicles
- Objects

The participant must arrange the feature maps from the simplest representation to the most complex representation.

Expected Outcome

Participants understand hierarchical feature extraction.

The progression can be represented as:

Pixels → Edges → Textures → Shapes → Object Parts → Objects

This demonstrates one of the most powerful characteristics of CNNs: the network automatically learns increasingly meaningful representations as the depth increases.

Level 3 – Regularization Rescuer

Challenge

In this level, participants are presented with a CNN that performs very well on training data but poorly on unseen test data. They must identify the problem and select suitable regularization methods.

Problem: Overfitting

Overfitting occurs when a model learns the training data too closely, including noise and unnecessary details. As a result, it performs poorly on new data.

For example:

Training Accuracy = 98%

Validation Accuracy = 72%

This large difference indicates possible overfitting.

Learning Activity

Participants choose appropriate techniques to improve generalization.

Common regularization methods include:

Dropout

Dropout randomly disables some neurons during training. This prevents the network from becoming overly dependent on specific neurons.

Data Augmentation

Images can be modified through operations such as:

- Rotation
- Flipping
- Cropping
- Scaling
- Translation

This creates additional variations of training images.

L2 Regularization

L2 regularization adds a penalty to large weights, encouraging the network to learn simpler models.

Early Stopping

Training can be stopped when validation performance stops improving, reducing the possibility of overfitting.

Expected Outcome

Participants learn how to identify overfitting and select suitable techniques to improve CNN generalization.

Level 4 – CNN Architecture Race

Challenge

Participants compare AlexNet and ResNet for different image-classification problems and select the more appropriate architecture.

Learning Activity

Different scenarios are provided.

For a relatively simple image-classification problem with limited computational resources, a simpler architecture may be preferred.

For a complex image-recognition task requiring a deeper network, ResNet may be more appropriate because residual connections help train very deep networks.

Participants compare the architectures based on:

Feature	AlexNet	ResNet
Architecture	Deep CNN	Very deep residual CNN
Key innovation	ReLU and deep CNN design	Residual/skip connections
Depth	Relatively smaller	Can be very deep
Skip connections	No	Yes
Training deep networks	More difficult	Easier
Major use	Image classification	Image classification and computer vision
Main advantage	Historically important CNN architecture	Effective deep architecture

Expected Outcome

Participants learn that architecture selection depends on the problem, dataset, computational resources, and required performance.

Level 5 – CNN Application Master

Challenge

The final level requires participants to design a CNN-based solution for a real-world problem.

Each participant or team receives an application scenario such as:

- Medical image analysis
- Face recognition
- Object detection
- Autonomous driving
- Security surveillance
- Agricultural disease detection

Example: Medical Image Analysis

Suppose a hospital wants to automatically classify chest X-ray images into normal and disease-affected categories.

A possible CNN system can contain:

X-ray Image → Preprocessing → Convolution → ReLU → Pooling → Convolution → Pooling → Flatten → Fully Connected Layer → Classification

The system can learn visual patterns associated with different conditions.

Example: Autonomous Driving

A CNN can process camera images from a vehicle and identify:

- Road lanes
- Vehicles
- Pedestrians
- Traffic signs
- Obstacles

For a more complex system, a suitable deep architecture can be selected based on the required accuracy and computational resources.

Example: Face Recognition

A CNN can extract facial features such as:

- Eyes
- Nose
- Mouth
- Face shape
- Local facial patterns

These features can then be used for identification or verification.

Architecture Selection

Participants must justify their architecture based on:

- Dataset size
- Image complexity
- Required accuracy
- Computational resources
- Training time
- Risk of overfitting
- Real-world requirements

Expected Outcome

This final level demonstrates whether participants can transfer theoretical CNN knowledge to a practical problem. The learner must identify the problem, design an appropriate CNN pipeline, select an architecture, and explain why the selected architecture is suitable.

Game Scoring and Progression

The activity can use a points-based scoring system to make learning more engaging.

Each level can carry 20 points, giving each round a maximum of 100 points.

Round Levels Maximum Score

Round 1 5 100

Round 2 5 100

Total 10 200

Additional bonus points can be awarded for:

- Correct answers without hints.
- Fast problem solving.
- Correct parameter calculations.
- Successful architecture design.
- Strong justification of real-world applications.

A leaderboard can be maintained to display participant rankings.

Suggested achievement levels are:

- 0–79: CNN Beginner
- 80–119: CNN Explorer
- 120–159: CNN Practitioner
- 160–179: CNN Specialist
- 180–200: CNN Master

This scoring mechanism encourages students to actively participate and improves motivation.

Overall Learning Outcomes

After completing both rounds, participants should be able to:

1. Explain the basic architecture of a Convolutional Neural Network.
2. Identify the functions of convolution, pooling, flatten, and fully connected layers.
3. Explain the purpose of filters and kernels.
4. Calculate convolution output dimensions.
5. Understand stride and padding.
6. Explain parameter sharing in CNNs.
7. Calculate the number of parameters in convolutional layers.

8. Interpret CNN feature maps.
9. Explain hierarchical feature extraction.
10. Identify and address overfitting.
11. Explain regularization techniques such as dropout, data augmentation, L2 regularization, and early stopping.
12. Compare AlexNet and ResNet.
13. Explain the importance of residual connections.
14. Select an appropriate CNN architecture for a given application.
15. Design a CNN-based solution for real-world problems.

Conclusion

The two-round game-based learning activity provides an interactive approach to understanding Convolutional Neural Networks. Instead of learning CNN concepts only through theoretical explanations, participants actively solve problems, calculate parameters, identify filters, debug architectures, analyze feature maps, and design real-world solutions.

Round 1, CNN Architecture Builder, establishes the foundation by teaching CNN layers, filters, parameter sharing, architecture design, and the differences between AlexNet and ResNet. Round 2, CNN Feature Detective, builds upon this knowledge by focusing on convolution operations, feature maps, regularization, architecture selection, and practical applications.

The progression from basic concepts to real-world problem solving makes the activity suitable for students learning deep learning and computer vision. By completing all ten levels, participants gain both conceptual understanding and practical reasoning skills. The game also demonstrates how CNNs transform raw image data into meaningful features and use those features to make intelligent predictions.

Overall, the activity makes CNN learning more engaging, competitive, and application-oriented while helping learners develop the knowledge required to understand and design modern computer vision systems.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Conceptual Clarity	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Problem-Solving Ability	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design & Application	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Teamwork & Game Participation	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

